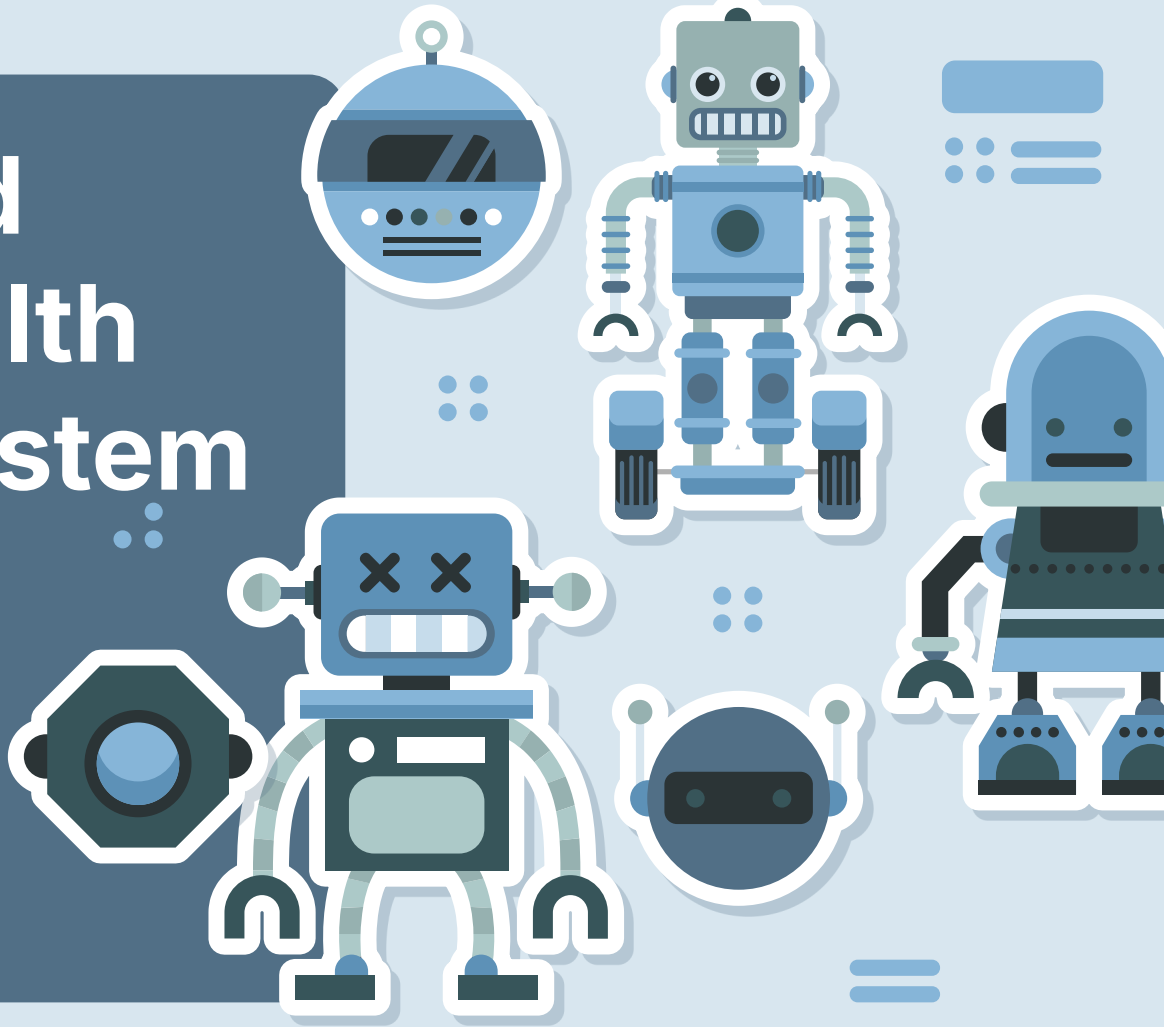
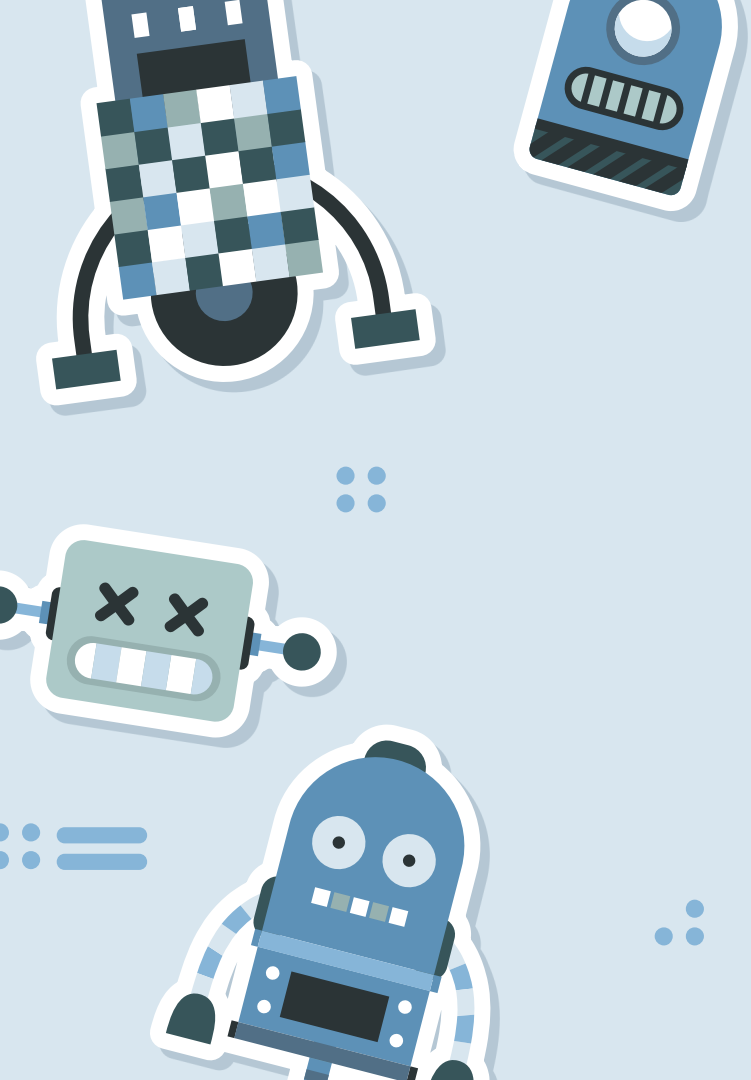
NAO-Based Mental Health Support System

Team 5: Aryanna Aneiros
Caroline Sant'ana Morvan

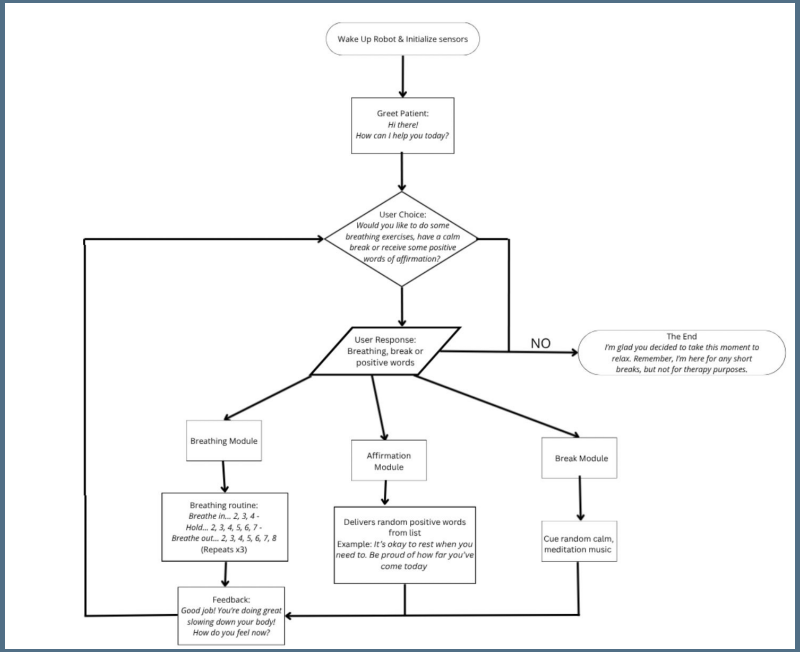


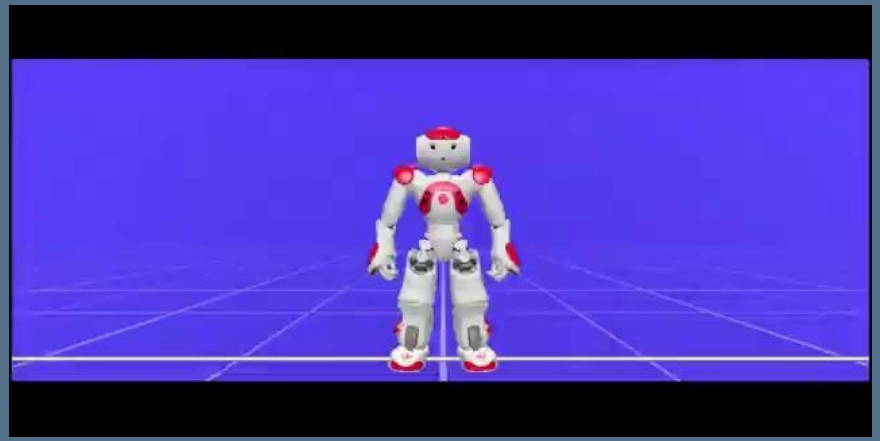


Overview

- NAO as a socially assistive tool to help reduce user stress levels & promote emotional well-being
- 4 total modules
- Main controllers: greets user and prompts selection of one of three interaction modules available
- Stress-relief activities: guided breathing, positive words of affirmation & calm break modules
- Feedback loop: NAO checks in with user, options are to either end interactive session or choose another module

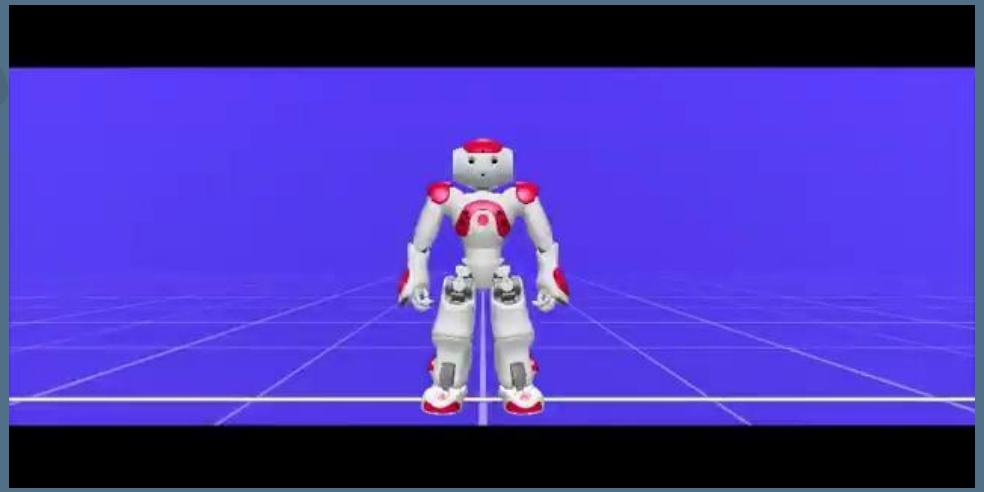
Flowchart


Wake Up Robot & Initialize sensors
Greet Patient:
Hi there!
How can I help you today?
User Choice:
Would you like to do some breathing exercises, have a calm break or receive some positive words of affirmation?
User Response:
Breathing, break or positive words
NO
The End
I'm glad you decided to take this moment to relax. Remember, I'm here for any short breaks, but not for therapy purposes.
Breathing Module
Affirmation Module
Break Module
Breathing routine:
Breathe in... 2, 3, 4 -
Hold... 2, 3, 4, 5, 6, 7 -
Breathe out... 2, 3, 4, 5, 6, 7, 8
(Repeats x3)
Delivers random positive words from list
Example: It's okay to rest when you need to. Be proud of how far you've come today
Cue random calm, meditation music
Feedback:
Good job! You're doing great slowing down your body!
How do you feel now?

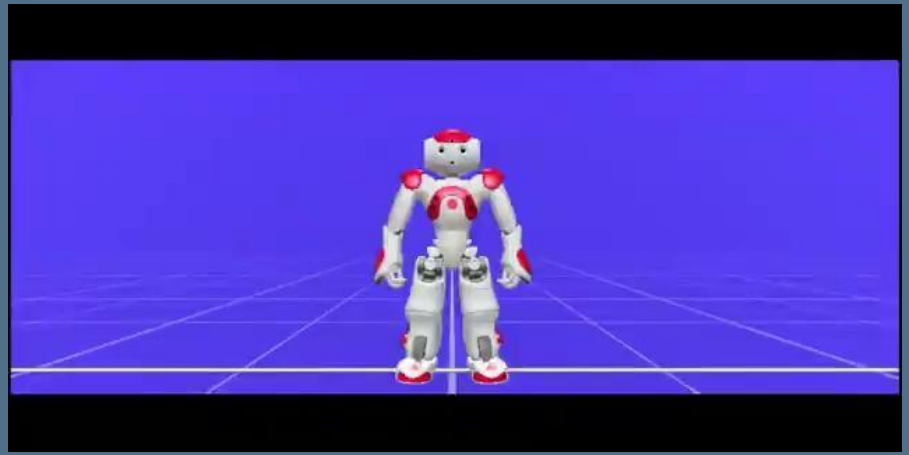


NAO main controller demonstration

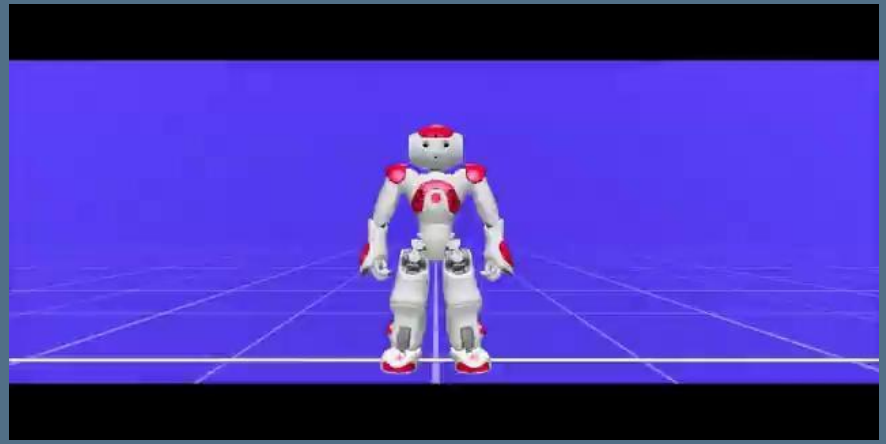
(tts to describe how speech recognition should work)



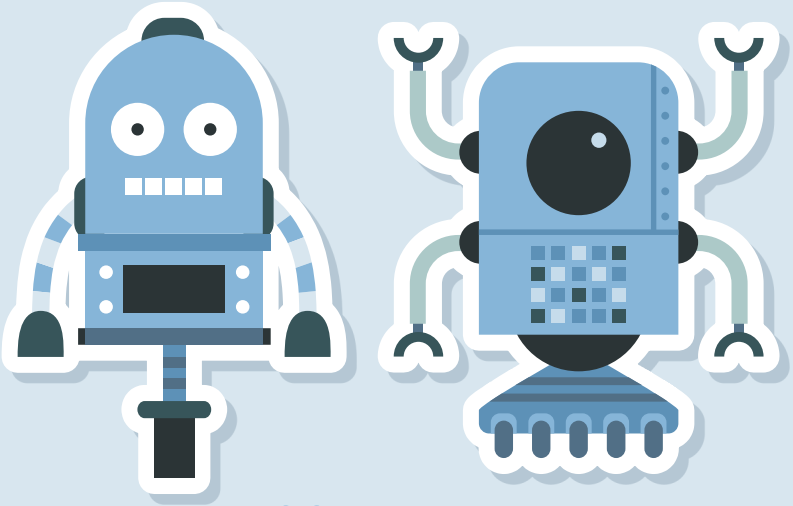
NAO-guided 4–7–8 breathing sequence



NAO-guided calm break sequence



NAO-guided positive words of affirmation sequence



Conclusion

- Developed & tested activity modules (breathing, calm break, and affirmation)
- Demonstrated at modular level a NAO-based stress-relief system
- Simulation limited speech recognition, audio and full interaction testing